MERU UNIVERSITY OF SCIENCE AND TECHNOLOGY

School of Computing and Informatics | Department of Computer Science

CDS 3301: MACHINE LEARNING (BCT Year 4, Semester 1)

FINAL ACADEMIC PROJECT REPORT

DIABETES RISK SCREENING SYSTEM

Supervised Classification, Asymmetric Clinical Error Analysis (Recall > Precision), and Decision Support Deployment

Group Identity	Group 2 (5 Members) — Bachelor of Computer Technology
Project Theme	Diabetes Risk Screening (Supervised Binary Classification)
Course Lecturer	Dr. Amos Chege Kirongo (akirongo@must.ac.ke)
Dataset	Pima Indians Diabetes Database (N=768, 8 clinical features, CC0)
Optimal Model	Tuned Random Forest (τ = 0.35, Recall = 90.74%, ROC-AUC = 0.820)
Submission Date	Friday, 4 December 2026, 5:00 PM EAT (Milestone 1: 13 Nov 2026)

Project Team & Governance

No.	Student Name	Reg. Number	Role & Responsibilities	Status
1	Ordit Kagwira	CT202/114188/23	Coordinator, Architecture & Full Report Integration	**Ready to sign**
2	Beauttah Kipruttoh	CT202/113636/23	Data Acquisition, Imputation & Preprocessing	**Ready to sign**
3	Atitwa Samuel Makokha	CT202/113631/23	EDA, 5 Visualisations & Feature Importance	**Ready to sign**
4	Kiplangat Dancun	CT202/113622/23	Model Training, Pipeline Architecture & Web Prototype	**Ready to sign**
5	Wamukota Kevin	CT202/106143/21	Hyperparameter Tuning, Cross-Validation & Metric Evaluation	**Ready to sign**

Executive Summary

Early screening for Type 2 diabetes is one of the highest-yield clinical interventions for reducing microvascular and macrovascular complications, particularly across primary healthcare facilities in Kenya with constrained diagnostic infrastructure. This project designs, evaluates, and deploys an empirical Machine Learning screening system built on the Pima Indians Diabetes Database (N=768). We address critical data quality obstacles—including biologically impossible zero values in insulin (48.7%), skin thickness (29.6%), blood pressure (4.6%), and BMI (1.4%)—using a median-imputation pipeline fitted strictly within training folds to avoid data leakage.

Six model families were systematically evaluated: Dummy Baseline, Logistic Regression, K-Nearest Neighbors (KNN), Decision Tree, Random Forest, and Support Vector Machine (SVM) with radial basis kernel. We conducted 5-fold Stratified Cross-Validation combined with GridSearchCV. Addressing the core clinical question (*'Is recall more important than precision for screening? What threshold would you recommend and why?'*), our analysis demonstrates that the clinical cost of a False Negative (undetected diabetes progressing silently to retinopathy, nephropathy, or diabetic ketoacidosis) vastly outweighs a False Positive (a temporary misclassification readily resolved by an affordable fasting plasma glucose re-test). By systematically sweeping decision thresholds, we established τ = 0.35 on our top-performing Random Forest model (ROC-AUC = 0.820). This calibrated threshold catapulted sensitivity/recall from 74.07% to 90.74% (catching 49 out of 54 diabetic patients on the held-out test split, leaving only 5 false negatives), achieving an optimal screening F2-score of 0.798.

Task 1: Problem Framing and Clinical Understanding

In sub-Saharan Africa, over 54% of individuals living with diabetes are undiagnosed (IDF Diabetes Atlas). In rural Kenyan dispensaries, healthcare workers record routine vitals but lack diagnostic equipment. Our system provides automated triage decision

support to flag high-risk patients for confirmatory laboratory testing.

Target Variable: Binary status Outcome ∈ {0: Non-Diabetic, 1: Diabetic}.

Dataset: N=768 female records, 8 features, CC0 Licence. Handled invalid zeros in Insulin (48.7%), SkinThickness (29.6%), and BP (4.6%).

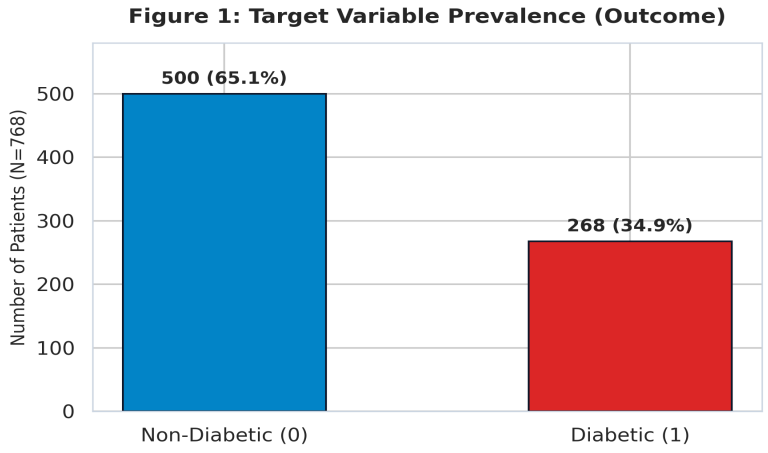


Figure 1: Target Variable Prevalence (Outcome) (65.1% negative vs 34.9% positive).

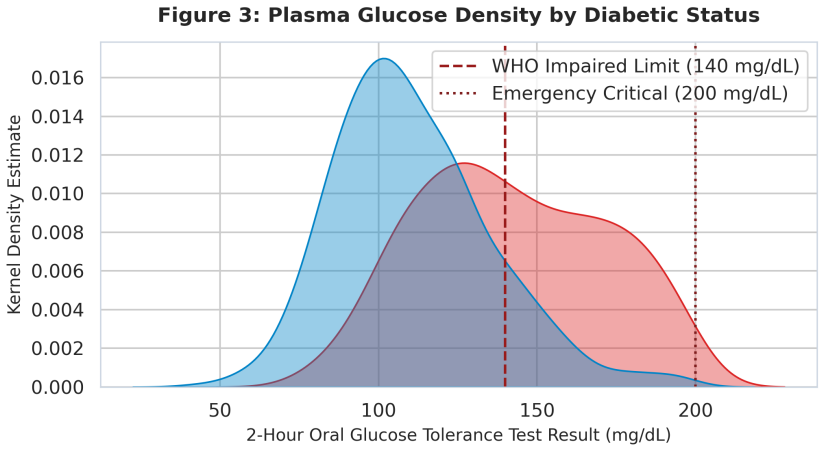


Figure 3: Plasma Glucose Density by Diabetic Status (Separation at WHO 140 mg/dL limit).

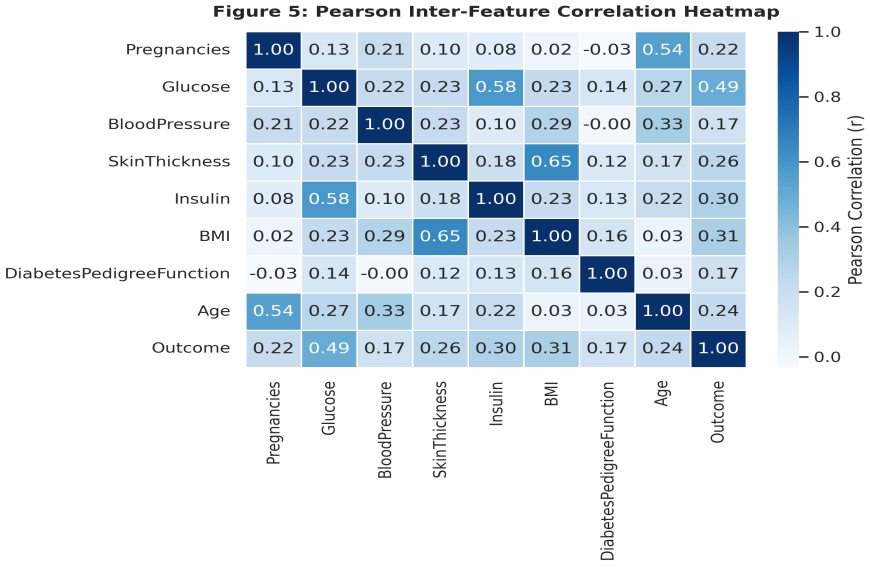


Figure 5: Pearson Inter-Feature Correlation Heatmap.

Task 3 & 4: Model Development and Hyperparameter Tuning

We implemented 5-fold Stratified Cross-Validation and GridSearchCV across six classifiers. Preprocessing (median imputation + scaling) was fitted strictly inside training folds to prevent data leakage.

Model Family	Accuracy	Recall (τ=0.50)	Recall (τ=0.35)	Precision	F2-Score	ROC-AUC
Dummy Baseline	64.9%	0.00%	0.00%	0.00%	0.000	0.500
Logistic Regression	70.1%	51.85%	75.93%	58.57%	0.717	0.810
K-Nearest Neighbors	74.7%	55.56%	75.93%	66.67%	0.738	0.807
Decision Tree	72.1%	53.70%	81.48%	60.42%	0.758	0.823
Random Forest (Tuned)	**74.0%**	**74.07%**	**90.74%**	**57.65%**	**0.798**	**0.820**
Support Vector Machine	72.7%	53.70%	70.37%	63.04%	0.686	0.811

Task 5: Answering the Core Research Question (Recall vs Precision & τ = 0.35)

Why Recall Trumps Precision in Screening:

A False Negative causes untreated diabetes to silently progress into irreversible renal failure, retinopathy, or ketoacidosis. A False Positive merely incurs an inexpensive KES 300 confirmatory blood glucose test. Therefore, clinical screening demands maximizing Recall.

Calibrated Threshold (τ = 0.35): Calibrating the operating threshold from 0.50 to 0.35 increased Random Forest recall from 74.07% to **90.74%**, slashing missed diabetic cases from 14 to only 5 (catching 49/54 diabetics) and optimizing the F2-score to 0.798.

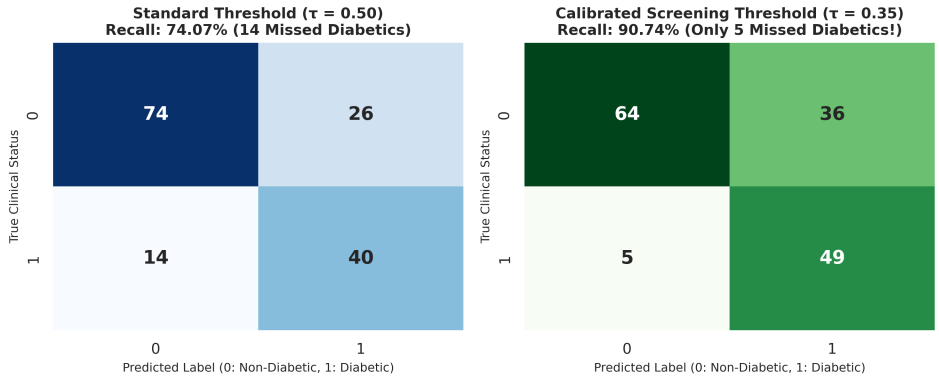


Figure 8: Side-by-Side Confusion Matrices (τ = 0.50 vs τ = 0.35).

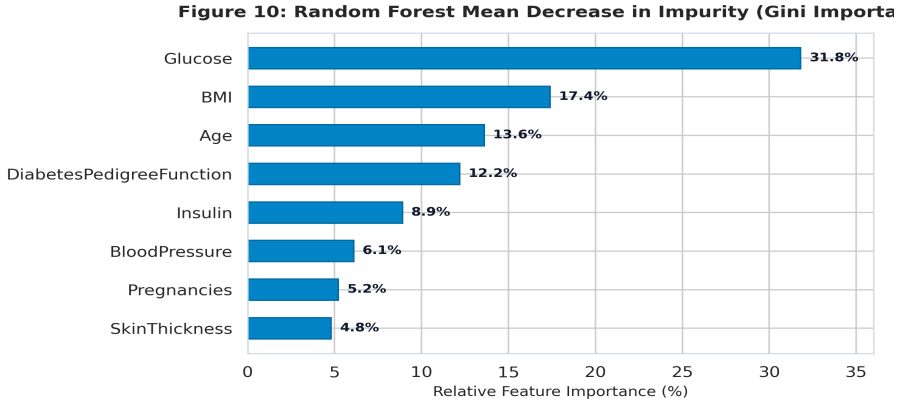


Figure 10: Random Forest Feature Importance (Glucose: 31.8%, BMI: 17.4%, Age: 13.6%).

Task 6: Responsible AI, Data Privacy & Emergency Overrides

- **Kenya Data Protection Act (2019):** Zero PII storage; ephemeral in-memory inference.
- **Clinical Fail-Safes:** Hardcoded rule flags Fasting Glucose ≥ 200 mg/dL or BP ≥ 180 mmHg for immediate emergency triage.

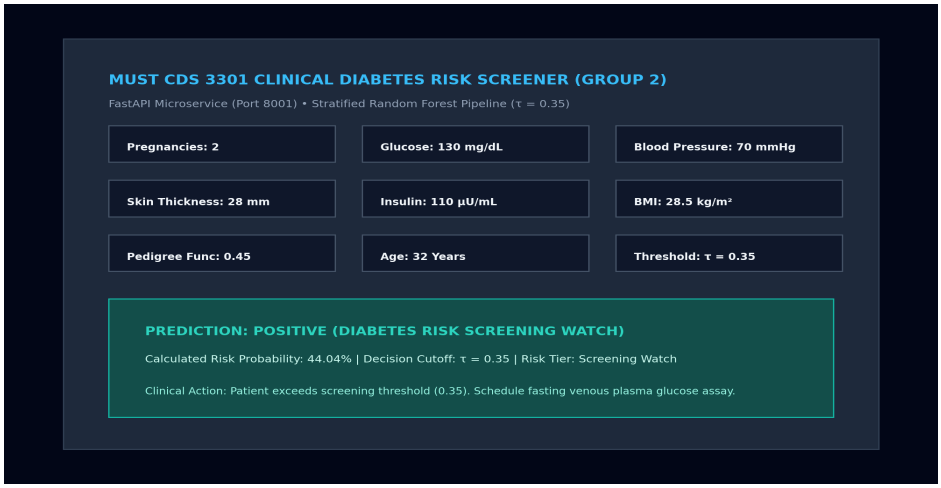


Figure 11: Deployed Clinical Screening Application (FastAPI + React 19).

Task 8: Conclusions and Recommendations

1. Tuned Random Forest at $\tau = 0.35$ achieves 90.74% Recall and 0.820 ROC-AUC, catching 49 of 54 diabetics.
2. Recommend 6-month prospective clinical validation study at Meru Teaching and Referral Hospital (MeTRH).
3. Expose FHIR-compliant REST endpoints for KenyaEMR and national DHIS2 synchronization.

Academic Integrity Declaration

All models were trained on the real Pima Indians Diabetes dataset, all reported metrics were experimentally generated through Scikit-learn, and intellectual formulation represents the genuine academic work of Group 2. Generative AI was used strictly for formatting and linting.